

Lab 3 - 11247123 Dao Phat Nhat

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Windows PowerShell
h-account) (2026.9.10)
Requirement already satisfied: py_ecc>=5.2.0 in C:\Users\admin\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313\site-packages (from eth-keyfile>=0.10->eth-account) (8.0.0)
Requirement already satisfied: pycryptodome<4,>=3.6.6 in C:\Users\admin\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313\site-packages (from eth-keyfile>=0.10->eth-account) (3.23.0)
Requirement already satisfied: typing_extensions>=4.5.0 in C:\Users\admin\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313\site-packages (from eth-typing>=3.0.0->eth-abi>=4->eth-account) (4.16.0)
Requirement already satisfied: eth-hash>=0.3.1 in C:\Users\admin\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313\site-packages (from eth-utils>=6->eth-account) (0.8.0)
Requirement already satisfied: cytoolz>=0.10.1 in C:\Users\admin\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313\site-packages (from eth-utils>=6->eth-account) (1.1.0)
Requirement already satisfied: annotated-types>=0.6.0 in C:\Users\admin\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313\site-packages (from pydantic>=2->eth-account) (0.8.0)
Requirement already satisfied: pydantic-core==2.46.5 in C:\Users\admin\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313\site-packages (from pydantic>=2->eth-account) (2.46.5)
Requirement already satisfied: typing-inspection>=0.4.2 in C:\Users\admin\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313\site-packages (from pydantic>=2->eth-account) (0.4.4)
Requirement already satisfied: toolz>=0.8.0 in C:\Users\admin\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313\site-packages (from cytoolz>=0.10.1->eth-utils>=6->eth-account) (1.1.0)
PS C:\WINDOWS\System32> python -c "import eth_account; print('eth-account installed')"
eth-account installed
PS C:\WINDOWS\System32>
```

```
PS D:\hoc\Giao Trinh\Giao Trinh\Stats> python .\pow_demo.py
hash 1: c42dafbbe164b917016b4c3491ee7445e6c01fae0524a83f96c2ece8720b2d78
hash 2: 3b467304922f266e0331cc5e1598ec58657f48b5a9e2587763260244a30f40c7
=1 nonce= 5 time=0.000s
=2 nonce= 61 time=0.000s
=3 nonce= 61 time=0.000s
=4 nonce= 14229 time=0.022s
=5 nonce= 55980 time=0.075s
```

```
CHECK 1 (all 8 proofs valid): OK
CHECK 2 (proof length == 3): OK
CHECK 3 (tampered leaf fails): OK
CHECK 4 (odd count works): OK
```

```
PS D:\hoc\Giao Trinh\Giao Trinh\Stats> python ".\signature_demo.py"
=== ACCOUNT ===
Original address: 0xbEa6FFAbE4454d9C07b7f5850754680533f9A4e5

=== ORIGINAL MESSAGE ===
Message: I attended Session 3 / Toi da hoc Buoi 3

=== SIGNATURE ===
r: 0x16f59c558b14ad1757c3594048034fd272f008886abb59748b5cd15906261f23
s: 0x73b5dd0648d4d846f17e77b39be233096d6bed802ed61947b4afb854d825d16
v: 27
Signature: 16f59c558b14ad1757c3594048034fd272f008886abb59748b5cd15906261f23073b5dd0648d4d846f17e77b39be233096d6bed802ed61947b4afb854d825d161b
Two signatures are identical: True

=== RECOVERY ===
Recovered address: 0xbEa6FFAbE4454d9C07b7f5850754680533f9A4e5
Recovered matches original: True

=== TAMPERED MESSAGE ===
Tampered message: I attended Session 3 / Toi da hoc Buoi 4
Tampered recovered address: 0x72f7a035f4408e21219148C3384e0392Fe8410E1
Tampered matches original: False
```